natural tendency to give them an equal number of little brothers as well. From the point of view of a worker, the chance of any one brother containing a particular one of her genes is only $\frac{1}{4}$. Therefore, if the queen were allowed to produce male and female reproductive offspring in equal proportions, the farm would not show a profit as far as the workers are concerned. They would not be maximizing the propagation of their precious genes.

Trivers and Hare realized that the workers must try to bias the sex ratio in favour of females. They took the Fisher calculations on optimal sex ratios (which we looked at in the previous chapter) and re-worked them for the special case of the Hymenoptera. It turned out that the stable ratio of investment for a mother is, as usual, 1:1. But the stable ratio for a sister is 3:1 in favour of sisters rather than brothers. If you are a hymenopteran female, the most efficient way for you to propagate your genes is to refrain from breeding yourself, and to make your mother provide you with reproductive sisters and brothers in the ratio 3:1. But if you *must* have offspring of your own, you can benefit your genes best by having reproductive sons and daughters in equal proportions.

As we have seen, the difference between queens and workers is not a genetic one. As far as her genes are concerned, an embryo female might be destined to become either a worker, who ‘wants’ a 3:1 sex ratio, or a queen, who ‘wants’ a 1:1 ratio. So what does this ‘wanting’ mean? It means that a gene that finds itself in a queen’s body can propagate itself best if that body invests equally in reproductive sons and daughters. But the same gene finding itself in a worker’s body can propagate itself best by making the mother of that body have more daughters than sons. There is no real paradox here. A gene must take best advantage of the levers of power that happen to be at its disposal. If it finds itself in a position to influence the development of a body that is destined to turn into a queen, its optimal strategy to exploit that control is one thing. If it finds itself in a position to influence the way a worker’s body develops, its optimal strategy to exploit that power is different.

This means there is a conflict of interests down on the farm. The queen is ‘trying’ to invest equally in males and females. The workers are trying to shift the ratio of reproductives in the direction of three females to every one male. If we are right to picture the workers as the farmers and the queen as their brood mare, presumably the workers will be